RIN RAY

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CRC Deformations-Geometry-Structure Postdoctoral Fellow, Universität Münster

ACADEMIC EMPLOYMENT

2023–Present **CRC Postdoctoral Fellow**; Universität Münster; Arithmetic and Homotopy Theory Working Group, with Thomas Nikolaus and Christopher Deninger
- chromatic homotopy theory · the Lubin–Tate action · moduli stacks of curves · arithmetic geometry

2015 **Visiting Researcher**; Santa Fe Institute of Complex Systems; Santa Fe, NM
- multiscale physics modelling · directed type systems

2013 **Computational Chemistry Researcher**; George Mason University, VA; advised by Dimitrios Papaconstantopoulos
- superconducting transition temperature · electronic-structure calculation · augmented plane wave method · composite materials

2013 **Automated Computational Semantics & Machine Learning Researcher**; George Mason University; Fairfax, VA
- probabilistic lexicon induction · natural language processing · decipherment

2012 **Human-Computer Interaction Researcher** (Autonomous Robotics); George Washington University; Washington D.C.
- proprioceptive-only control under graceful degradation · human–robot interaction

2012 **Autonomous Aerial Vehicle Researcher**; George Mason University
- simultaneous localization and mapping (SLAM) · motion planning · visual perception for unmanned aerial vehicles in novel environments

2011 **Research Assistant in Materials Science Lab**; Mary Baldwin College; Staunton, VA
- memristive devices · resistive switching

EDUCATION

2017–2023 **Northwestern**; PhD in Maths; advised by Paul Goerss
- *A Geometric Model of Higher K-theories at Height $h=p^{k-1}(p-1)$ via Families of Ramified Curves*

2015–2017 **UChicago**; Divisional Masters Program in Maths; advised by Peter May and Kazuya Kato
- *Calculating $\pi_*(tmf)$ at the prime 2*
- *Models of Formal Group Laws of Every Height*

2012–2013 **George Mason University**; B.S. in Computational Physics (graduated winter 2013, at 16); advised by Dimitrios Papaconstantopoulos
- *Contextual Machine Learning through the Analysis and Chunking of Partially Translated Grade 2 Braille*
- *Predicting Superconductivity Transition Temperature via the Augmented Plane Wave Model*

2010–2012 **Mary Baldwin College**; Program for the Exceptionally Gifted: Mathematics, Computer Science, and Physics; advised by John Ong

PUBLICATIONS

2022 **Automorphisms of Abelian Varieties and Principal Polarizations**; joint with D. Lee; *Rendiconti del Circolo Matematico di Palermo*, Series 2, vol. 71, pp. 483–494; arXiv:1811.07007

2020 **Towards Directed Collapsibility**; joint with R. Belton, R. Brooks, S. Ebli, L. Fajstrup, B. T. Fasy, N. Sanderson, E. Vidaurre; *Advances in Mathematical Sciences*, vol. 21, pp. 255–271; arXiv:1902.01039

PREPRINTS

Sept 2026	**On the Chromatic Splitting Conjecture in Coheight 1**; joint with T. Barthel, L. Mann, A. Senger, T. Schlank, J. Weinstein, X. Zhou; on Jacquet–Langlands and homotopy theory
Sept 2025 UPDATED SEPT 2026	**Moduli Stacks of *G*-Curves in Homotopy Theory at $h = p-1$**; arXiv:2509.23428
July 2025	**Towards the $p = 3$ Kervaire Invariant Problem**; joint with E. Belmont; the E_2-page for the homotopy fixed points spectral sequence computing $\pi_*(E_6^{hC_9})$; arXiv:2507.10157
July 2025	**Modeling Group Actions on Stacks (Especially the Lubin–Tate Action)** under construction ; arXiv:2507.00309
2018	**A Global Crystalline Period Map**; joint with M. Neaton and A. Pieper; from the Arizona Winter School; arXiv:1911.08615

IN PROGRESS

Zeta Functions & Geometric Langlands	***L*-genera and localizations in *K*-theory**; joint with D. Berwick-Evans and N. Pacheco-Tallaj
Moduli Stacks & Geometric Langlands	**Bounded modifications of *G*-curves and homotopy theory at $h = p^{k-1}(p-1)$**
Zeta functions	**Syntomic cohomology of ring spectra and a *T*(*h*)-local zeta function**; joint with G. Angelini-Knoll
Zeta functions	**Toward categorifying the relationship of symplectic *L*-functions and Reidemeister torsion**; Riemann–Roch for Reidemeister torsion in *K*-theory and *L*-theory
Moduli stacks	**The eigenvalues of Frobenius of Artin–Schreier–Witt curves are Gauss sums**; new families of Newton strata in the Torelli locus
Zeta functions	**All Bernoulli numbers in homotopy theory are shifts** on hiatus ; joint with A. Mejia and N. Riggenbach; connecting Kervaire–Milnor to Quillen–Lichtenbaum

FELLOWSHIPS

2022	Nominated for **Presidential Fellowship**; Northwestern University
2017	Awarded **NSF GRFP Fellowship** (103K + 36K)
2014–2016	Awarded **Thiel Fellowship** (100K); mathematics and assistive technology; mentor Edward Frenkel

AWARDS

2022	**Gelfand Prize**, Northwestern University; for outstanding progress in one's own research and extraordinary contributions to the research goals of the graduate community
2013	**Outstanding Rising Senior Award** in Computational and Data Sciences, SPACS

SERVICE, TEACHING AND OUTREACH

Winter 2025	ORGANIZER **Linearity in *p*-adic Lie Theory**; Homotopy and Arithmetic joint working group
Spring 2025	ORGANIZER ***K*-Theoretic Poitou–Tate Duality**; Homotopy and Arithmetic joint working group
2025	ORGANIZER TEACHER **"Mathe trifft Kunst" (Math Meets Art)**; activity designer and co-organizer; hands-on mathematics workshops for children, Universität Münster
Summer 2024	ORGANIZER **Young Topologists Meeting**
2024	PANELIST **Queer in Math Day**; MPI Leipzig
Winter 2024	ORGANIZER **Euler Characteristics and Zeta Functions**; homotopy working group

Winter 2023	ORGANIZER **K-Theory of the Integers**; Homotopy and Arithmetic joint working group
2022–2023	FOUNDER **Mental Health Math Buddy Program** ongoing; Northwestern. Pairs first- and second-year graduate students with older graduate students, meeting once a quarter for mental health check-ups.
2022	FUNDING Established the **Fund for Work–Life Activities for Graduate Students**; Northwestern University
Winter 2022	ORGANIZER **Stop Staring and Compute!**; interactive methods of computation in number theory and algebraic geometry
2021–2023	MENTOR **TGS Diversity Peer Mentorship Program**; Northwestern. Upper-year graduate students help first-year PhD students from diverse backgrounds in other subjects acclimate, providing regular check-ins, encouragement and support.
2021–2023	MENTOR **Causeway Postbaccalaureate Program**; Northwestern. Mentoring students from diverse backgrounds through a bridge programme preparing them for graduate school.
2021–2023	MENTOR TEACHER **Directed Reading Program**; Northwestern. Weekly one-on-one work with an undergraduate on a topic of their choice, ending in a talk to their peers.
2020–2023	VOLUNTEER **Evanston Animal Shelter**; fostered special-needs adult cats and raised neonatal kittens.
Winter 2020	ORGANIZER **Moduli Spaces**; alternating algebraic and analytic
Spring 2019	ORGANIZER **Applications of Topos Theory**; homotopy working seminar
Winter 2019	ORGANIZER **TamagaWHAT**; on the proof of the Tamagawa conjecture for function fields
Winter 2019	ORGANIZER **Formal Moduli Problems**; homotopy working seminar
Fall 2018	ORGANIZER **Geometry for Prime Addicts**; *p*-adic geometry toward Scholze's proof of the monodromy weight conjecture
Fall 2018	ORGANIZER **Factorization Homology**; homotopy working seminar
2016–2017	TEACHER **Math Circles**; weekly volunteering with children aged 6–12, South Side of Chicago
2015–2017	TEACHER MENTOR **Research Experience for Undergraduates**; University of Chicago. Summer programme with weekly lectures and interaction, guiding each undergraduate through a targeted individual research project.
2014	FOUNDING CONTRIBUTOR **Topos House** ongoing; a still-active house in South San Francisco for scientific and artistic collaboration, especially between mathematicians and biologists.

OTHER EMPLOYMENT

2014–2016	**Thiel Fellow** (Math and Assistive Tech Inventor); Thiel Foundation; Berkeley, CA • medical and assistive technology · independent mathematics study under Edward Frenkel
2013–2014	**Mouse Vocalization Specialist & Audio Processing Engineer**; Vium; Belmont, CA • computational bioacoustics · unsupervised classification of ultrasonic vocalizations (USVs) · quasi-real-time audio processing
2013	**Software Engineering Intern**; Cloudera; Palo Alto, CA • Hadoop · Hive · large-scale data processing

SCIENTIFIC WORKS

2015	**Simplifying Multiscale Modeling**; Santa Fe Institute of Complex Systems
2014	**A New Female-Female Mouse Vocalization Discovered via Unlabeled Machine Learning**; Vium
2014	**On the Detection and Prevention of Aggression in Lab Mice via Quasi-Real Time Analysis**; Vium
2013	**Nonprovisional patent application**; Robotic Mobility Assistive Wheelchairs, 90776.0001USP1 • robotic mobility · pressure injury prevention

SELECTED INVITED TALKS

2026	**Mod *p* Langlands and the Chromatic Splitting Conjecture**; Essen Number Theory Seminar
2024	**Moduli Stacks of Curves in Homotopy Theory**; Johns Hopkins Topology Seminar
2021	**Galois Theory of Spheres**; Front Range Number Theory
2020	**Explicit Arithmetic of Jacobians**; ICERM Workshop on Arithmetic Geometry, Number Theory, and Computation
2019	**Inverse Galois Theory in the Homotopy Groups of Spheres**; Princeton CCR Colloquium
2019	**Galois Theory and Galois Representations in Homotopy Theory**; Higher School of Economics, Moscow
2019	**An Introduction to the Classic Theory of *p*-Divisible Groups**; Oberwolfach Arbeitsgemeinschaft

CREATIVE WORKS

2026	***Impaction***, first play; staged in Los Angeles as part of *Nothing to See Here*, directed by Jacques Manjarrez
2025	***Endomortis***, comic; with Petra Flurin; crayolapunk surrealist comic about grief